

Qatar-1b

R-2763

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_160726 · created 2026-09-03T00:34:20+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457595.90747 +/- 0.00084 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.079 +/- 0.047
Transit depth (Rp/Rs)^2	0.63 +/- 0.74 [%]
Orbital Inclination (inc)	79.25 +/- 3.1
Airmass coefficient 1 (a1)	1.055 +/- 0.031
Airmass coefficient 2 (a2)	-0.042 +/- 0.031
Scatter in the residuals of the lightcurve fit is	2.9 %
Best Comparison Star	#1 - [221, 460]
Optimal Aperture	3.09
Optimal Annulus	5.87
Transit Duration (day)	0.017 +/- 0.024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.079 ± 0.047 vs 0.1463 (deviation 1.4σ)
Duration fit vs published	0.017 d vs 0.0692 d (deviation 2.13σ)
Mid-time O–C	1.3 min
Depth SNR	1.6
Residual scatter	2.9 %

Light curve

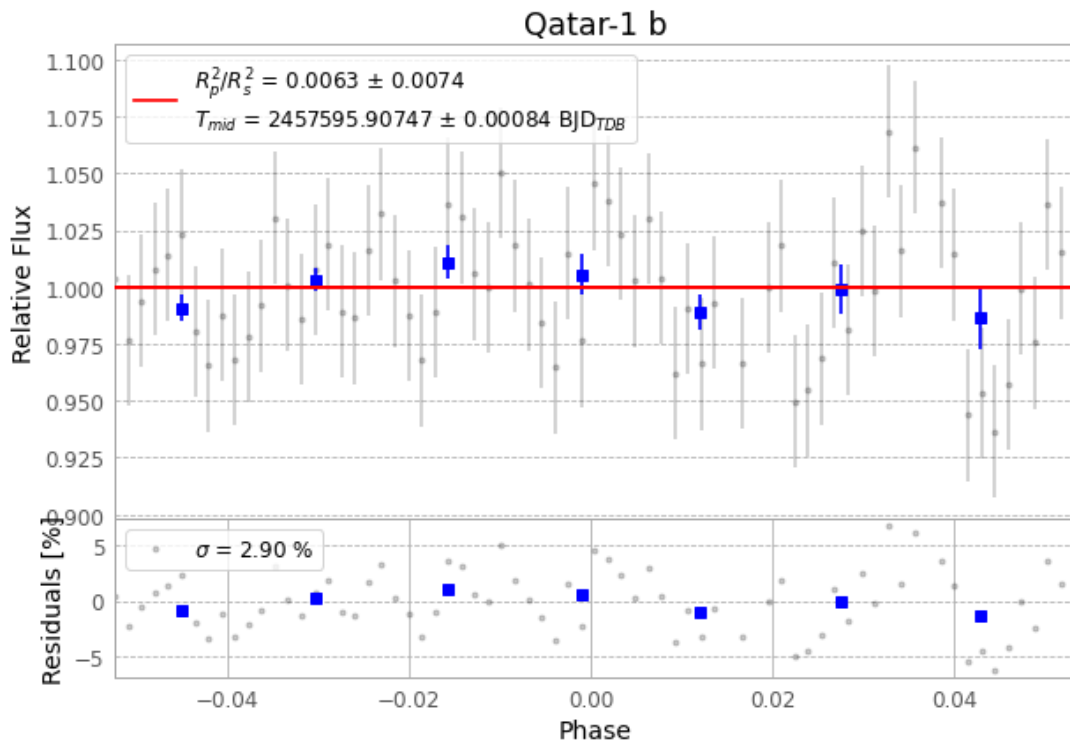


Figure 1 — FinalLightCurve_Qatar-1 b_2016-07-26.png (SHA-256 c0e879e1f927d3ef...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [257, 208], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0ab4df9f427d48ef...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

73 FITS frames (48 MB), Qatar-1160726075712.FITS → Qatar-1160726113312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-160726.log (1d834567ef69...), FinalLightCurve_Qatar-1 b_2016-07-26.png (c0e879e1f927...), FinalLightCurve_Qatar-1 b_2016-07-26.csv (0bf4c79648e8...)

Compute resources

Wall time 126.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-03 00:34Z.